INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Business Continuity and Disaster Recovery Assignment

GRC205 Incident Management and Business Continuity Week 21

Act as the Business Continuity and Disaster Recovery Coordinator for a fictional training institute whose main site becomes unavailable. Use the supplied data to assess business impact, set recovery objectives, select an affordable strategy, and demonstrate whether the proposed recovery sequence can meet business needs.

Assessment detail	Requirement
Assessment type	Individual assignment | Complete all six tasks
Name	Farida Baberin-Yor Beacher
Student ID	2025/GCR/12741
Submission Date	22 October 2026
Total marks and weighting	100 marks | 30% of the final course grade
Submission	One Word document through the E-Campus assessment page
Deadline	As published for this assessment in E-Campus
Suggested effort	12 to 16 hours

Learning outcomes

- Conduct a Business Impact Analysis (BIA) and prioritise critical functions.
- Justify recovery time and recovery point objectives using business requirements.
- Compare alternate sites and backup strategies against cost and capability.
- Coordinate continuity, disaster recovery, and incident response responsibilities.
- Test recovery assumptions and turn findings into measurable improvements.

Report requirements

Submit 3,000 to 3,500 words, excluding the cover, tables, reference list, and appendices. Present the six tasks in order. Include calculations, the BIA and recovery objective matrices, a strategy comparison and budget, a coordinated plan, a recovery schedule, an exercise log, and a performance dashboard.

The six tasks carry 95 marks. Report quality and referencing carry 5 marks. A score of 80 out of 100 contributes 24 out of the available 30 points to your final course score.

Northbridge Skills Institute — BC/DR Coordinator Report

Task 1: Business Impact Analysis

1.1 BIA Matrix — Five Functions

Function	Owner Role	Users Affected	Critical Resources	Dependencies	Minimum Acceptable Service	Priority
F1 Teaching and Assessment	Academic Coordinator	1,800 learners; 120 scheduled for 11:00 exam	LMS01, DB01, AUTH01, recovery network, 25 workstations, invigilators	AUTH01, DB01, power, network, exam content, invigilators	Supervised assessment delivered to 120 learners by 11:00; 40 via centre workstations, 80 via own devices; paper-based fallback with manual records if systems fail	1
F2 Fees and Finance	Finance Officer	Fee-paying learners, finance staff	FIN01, DB01, AUTH01, recovery network, external payment provider	AUTH01, DB01, external payment gateway (available), network	Read-only transaction log access; manual receipt issuance with sequential numbering; reconciliation queue established by 16:00	2
F3 Admissions and Registration	Admissions Officer	Prospective learners, registrations staff	REG01, DB01, AUTH01, recovery network	AUTH01, DB01, network	Manual application log with name, contact, programme, timestamp; supervisor countersignature; data entered into REG01 after restoration; deadline maintained at 18:00	3

Function	Owner Role	Users Affected	Critical Resources	Dependencies	Minimum Acceptable Service	Priority
F4 Staff Payroll	HR Manager	90 staff members	HR01, DB01, AUTH01, recovery network	AUTH01, DB01, bank submission portal, network	Verified payroll file from last confirmed run; no new changes processed without dual authorisation; bank submission file prepared offline, held pending system validation before 12:00 Day 1	4
F5 Records and Certificates	Records Officer	Learners and external requesters	FS01, AUTH01, recovery network	AUTH01, FS01, network	Queue requests received; no certificates issued without integrity check post-restoration; acknowledgement sent within 48-hour commitment window	5

Dependency risks identified:

1. **DB01 is a single shared dependency for F1 through F4.** Failure to restore DB01 blocks all four application systems simultaneously. No DB01 means no LMS01, FIN01, REG01, or HR01.
2. **AUTH01 gates all five functions.** Without the local directory restored, no application can authenticate users, regardless of network or application availability.
3. **Production and backup administration share one administrator identity.** If that identity is compromised or unavailable, access to both the live environment and the backup catalogue is at risk simultaneously. The offsite provider's access-independence controls are undocumented, meaning there is no verified separation between the attacker's reach (in a security scenario) and the backup copies.

1.2 Impact Calculations

Assumptions: Disruption begins at 08:00. Hourly rates apply continuously per function for hours in which that function remains unrestored. The examination rescheduling cost of NGN 120,000 is applied once if no acceptable assessment service is available by 11:00 — that is, if F1 remains unrestored for 3 hours or more (hours 1, 2, and 3 from 08:00). Hourly costs are treated as non-overlapping incremental costs.

8-Hour calculation (08:00 to 16:00):

Function	Hours Unrestored	Rate (NGN/hr)	Subtotal (NGN)	One-off Cost
F1	8	60,000	480,000	+120,000 (exam not available at 11:00)
F2	8	40,000	320,000	—
F3	8	20,000	160,000	—
F4	8	10,000	80,000	—
F5	8	5,000	40,000	—
Total			**1,080,000**	**+120,000 = NGN 1,200,000**

24-Hour calculation (08:00 Day 0 to 08:00 Day 1):

Function	Hours Unrestored	Rate (NGN/hr)	Subtotal (NGN)	One-off Cost
F1	24	60,000	1,440,000	+120,000
F2	24	40,000	960,000	—
F3	24	20,000	480,000	—
F4	24	10,000	240,000	—
F5	24	5,000	120,000	—
Total			**3,240,000**	**+120,000 = NGN 3,360,000**

Two limitations of these estimates:

First, the hourly rates assume a constant, uniform loss rate throughout the disruption. In practice, impact is rarely linear: F2's financial exposure concentrates around the 16:00 fee confirmation deadline, and F4's exposure peaks at the 12:00 Day 1 payroll submission. A flat hourly rate underrepresents peak-period risk and overestimates off-peak impact.

Second, the calculations capture only quantifiable financial loss. Learner welfare consequences — anxiety during a disrupted examination, loss of confidence in the institution, and potential appeals — do not appear in the NGN figures. Data integrity risk is also absent: if records are reconciled incorrectly after manual workarounds, the downstream cost of correction, regulatory scrutiny, or audit findings may exceed the direct hourly loss.

How deadlines affect priority:

The 11:00 examination is the earliest hard deadline and affects 120 learners with confirmed obligations. F1 therefore takes first priority regardless of its hourly rate relative to F2. F2's 16:00 deadline is the next fixed business constraint, followed by F3's 18:00 admissions close. F4's 12:00 Day 1 deadline creates overnight pressure but allows the longest window before irreversible harm. F5 has a 48-hour internal commitment rather than a regulatory or financial deadline, placing it last.

Technical restoration order versus business priority:

Business priority ranks F1 first, but F1 cannot be restored before the recovery network, AUTH01, DB01, and LMS01 are operational. The restoration sequence is therefore technically determined:

network first, then AUTH01, then DB01 (which requires security checks), then LMS01. Business priority cannot override infrastructure dependencies; it determines which application receives restoration effort once the shared infrastructure is available.

Task 2: Recovery Objectives and Backup Design

2.1 Definitions and Targets

Recovery Time Objective (RTO): The maximum duration from disruption to the point at which a function is available at its defined minimum service level.

Recovery Point Objective (RPO): The maximum acceptable data loss expressed as time before the disruption at which the recovered data state must be current.

Maximum Tolerable Downtime (MTD): The absolute outer limit beyond which the organisation suffers irreparable harm. MTD must exceed RTO; the difference between them constitutes the recovery buffer.

Function	RTO	RPO	MTD	Justification
F1	2 hours (by 10:00)	1 hour	3 hours (by 11:00)	Examination starts at 11:00; a 1-hour buffer allows validation and learner setup. MTD is hard at 11:00.
F2	6 hours (by 14:00)	2 hours	8 hours (by 16:00)	Fee confirmation due 16:00; buffer allows manual reconciliation if restoration is late.
F3	8 hours (by 16:00)	4 hours	10 hours (by 18:00)	Admissions close 18:00; manual log covers interim period.
F4	24 hours (by 08:00 Day 1)	8 hours	28 hours (by 12:00 Day 1)	Bank submission 12:00 Day 1; payroll file can be prepared offline and validated before submission.
F5	24 hours	24 hours	48 hours	48-hour internal service commitment; no hard external deadline.

The recovery buffer for each function is embedded in the gap between RTO and MTD. Recovery is declared only when the application is accessible, data integrity checks pass, and a responsible business owner confirms the service is usable — not merely when systems appear online.

2.2 Checkpoint Analysis

At 08:00 on Day 0, the available checkpoints and their data ages are:

Backup Item	Checkpoint Time	Hours Before Disruption	Covers
B01 DB01 full	23:00 Day -1	9 hours	Base for F1–F4 applications
B02 DB01 incremental chain	07:00 Day 0	1 hour	Latest DB01 recoverable point (requires B01 + all incrementals to 07:00)
B03 AUTH01 snapshot	00:00 Day 0	8 hours	Local directory and configuration
B04 FS01 full	23:00 Day -1	9 hours	F5 records and certificates

Resulting data-loss windows:

- **DB01 (F1–F4):** Latest recoverable point is 07:00. Data entered between 07:00 and 08:00 — one hour — is not captured in the offsite chain. This one-hour window must be reconciled from email logs, payment provider records, or paper records where they exist. This falls within the RPO targets for F2 (2 hours), F3 (4 hours), and F4 (8 hours), but is close to the F1 RPO of 1 hour. For F1, the assessment data gap is acceptable only if no examination transactions occurred between 07:00 and 08:00.
- **AUTH01 (all functions):** The snapshot is 8 hours old. Any directory changes between 00:00 and 08:00 — new accounts, password resets, group changes — are absent. This falls within most RPOs but must be documented and communicated to staff.
- **FS01 (F5):** The checkpoint is 9 hours old. Certificate or record activity between 23:00 Day -1 and 08:00 Day 0 is unrecovered. Given the 24-hour RPO for F5, this is acceptable, but a request queue from that window must be confirmed through alternative records.

What can and cannot be recovered:

Payments confirmed by the external payment provider between 07:00 and 08:00 can be reconciled against provider transaction logs. Manual paper registrations taken before the disruption can be entered post-restoration. Examination submissions or attendance records from the disruption period are not automatically available and cannot be assumed recoverable without corroborating evidence from invigilators or learners.

2.3 Backup Design

Target schedule:

Backup Type	Frequency	Retention	Location
DB01 full	Weekly (Sunday 23:00)	4 weeks	Encrypted offsite, separate from live environment
DB01 incremental	Every 2 hours during business hours; every 4 hours overnight	7 days	Encrypted offsite
AUTH01 snapshot	Every 4 hours	7 days	Encrypted offsite
FS01 full	Nightly (23:00)	4 weeks	Encrypted offsite
FS01 incremental	Every 4 hours	7 days	Encrypted offsite
Immutable offline copy	Weekly	12 weeks	Air-gapped or WORM-locked storage, separate credentials

Restoration chain for DB01: Restore B01 (full, Sunday 23:00), then apply each incremental in sequence through to the desired recovery point. Each link in the chain must pass a checksum verification before the next is applied. A broken or missing incremental invalidates all subsequent links.

3-2-1 implementation:

- Three copies: production, offsite encrypted copy, immutable offline copy.
- Two media types: disk-based offsite and offline tape or WORM storage.
- One copy geographically separate from the primary site.

Access and integrity controls:

The current arrangement — where production and backup administration share one identity — must be separated. Backup administration requires a dedicated service account with no interactive login rights to production systems. The offsite provider must document and test access independence: the ability to retrieve backups without credentials from the production environment. Encryption keys must be stored separately from backup data.

Restoration tests must confirm that a full DB01 chain, from B01 through the latest incremental, produces a consistent database that passes application-level integrity checks — not merely that files are present. NIST SP 800-34 requires that contingency plans include procedures to verify that backups are usable before a disaster occurs (NIST, 2010). Relying on offsite storage alone does not establish recoverability; an intact file that produces a corrupt database on restoration provides no protection.

Task 3: Recovery Strategy and Budget

3.1 Option Comparison

Criterion	Cold	Warm	Hot
Estimated first service	24–48 hours	3–5 hours	~1 hour
Data-loss window	Up to 24 hours	1–6 hours	Up to 15 minutes
F1 RTO met (2 hrs)	No	Marginal (3–5 hr estimate)	Yes
F2 RTO met (6 hrs)	No	Yes (if lower end of range)	Yes
F3 RTO met (8 hrs)	No	Yes	Yes
F4 RTO met (24 hrs)	Possibly (upper end)	Yes	Yes
F5 RTO met (24 hrs)	Possibly	Yes	Yes
F1 RPO met (1 hr)	No	Marginal	Yes
Corruption propagation risk	Low (data validated before restore)	Moderate (incremental chain)	Higher (near-real-time replication may carry corrupt state)
Staff needed Day 0	2 technicians + significant rebuild	2 technicians	Minimal switching

Cold package: Cannot meet any of the critical deadlines on Day 0. The 24-to-48-hour window means the 11:00 examination, 16:00 fee confirmation, and 18:00 admissions close are all missed without recovery. Eliminated for functions with same-day deadlines.

Warm package: The 3-to-5-hour estimate places service restoration between 11:00 and 13:00, which misses the F1 deadline under the upper estimate. However, the drill data in Task 5 shows actual warm site restoration of assessment service is feasible by approximately 10:45 — within the RTO. The warm option meets RPO for all functions using the B02 07:00 checkpoint. Corruption propagation risk is moderate but manageable through validation steps.

Hot package: Meets all RTOs and RPOs. Replication-lag of up to 15 minutes is within F1's 1-hour RPO. The main concern is that near-real-time replication can propagate a corrupt or malicious data state before it is detected. Documented rollback procedures and monitoring are essential.

3.2 Budget

Package	Base Cost	Independent Protected Backups	Training and Tests	Total Annual
Cold	600,000	400,000	200,000	**1,200,000**
Warm	1,500,000	400,000	200,000	**2,100,000**
Hot	2,400,000	400,000	200,000	**3,000,000**

All three packages fall within the NGN 3,000,000 annual limit. Cold leaves the largest budget surplus but cannot meet the operational requirements. Hot reaches exactly the limit with no contingency margin.

Recommendation: Warm package at NGN 2,100,000.

The warm package meets all RTOs given the drill evidence in Task 5, meets all RPOs using the existing B02 07:00 checkpoint, and leaves NGN 900,000 of the annual planning limit available for contingency, additional controls, or capacity adjustments. The hot package's main advantage — sub-hour restoration — is not required by any of the five functions, and its higher replication speed introduces corruption propagation risk without a corresponding business need at this scale.

Residual risks that must be accepted:

The warm package's estimated 3-to-5-hour window is an estimate, not a guaranteed SLA. The F1 11:00 deadline requires that the warm site be activated immediately at 08:00 and that the drill timing of approximately 165 minutes (to LMS01 availability) be achievable on the day. If any action takes longer than estimated, the examination deadline is at risk. The continuity procedure for F1 — supervised paper-based examination — must be maintained as a parallel fallback regardless of which package is selected.

The executive steering committee must formally accept the residual risk that the warm package cannot guarantee F1 restoration before the examination in all scenarios.

Investment reasoning:

The 24-hour disruption cost estimate is NGN 3,360,000 — more than the annual warm package cost of NGN 2,100,000. This comparison is illustrative only. A single-outage estimate cannot establish annual return on investment because the probability of a comparable disruption occurring in any given year is unknown, disruptions vary in duration and scope, and indirect costs such as reputational damage and regulatory attention are not captured in the hourly rates. Before approving expenditure, the organisation needs: supplier SLA documentation with penalty clauses, an independent restoration test result from the proposed warm site, evidence of access independence for the offsite backup provider, and a minimum of one tabletop exercise validating the recovery sequence.

Task 4: Business Continuity and Disaster Recovery Plan

4.1 Plan Scope and Activation

Scope: All five functions at Northbridge Skills Institute following any event that makes the main site unavailable or any critical system inaccessible for more than one hour.

Activation thresholds: The BC/DR plan activates when the site is confirmed inaccessible, when AUTH01 or DB01 is unavailable and cannot be restored within 30 minutes, or when facilities management confirms an outage of more than two hours.

Authorised decision maker: BC/DR Coordinator (primary). Deputy: Academic Coordinator. Executive sign-off required for expenditures above NGN 200,000 or for declaration of full site loss.

Alternative communication: Microsoft 365 mobile email is confirmed available. WhatsApp group for core team as secondary channel. Landline tree for staff without mobile internet.

4.2 RACI Matrix

Activity	Responsible	Accountable	Consulted	Informed
Plan activation	BC/DR Coordinator	Executive Sponsor	Facilities Manager	All staff
Temporary assessment delivery	Academic Coordinator	BC/DR Coordinator	Invigilators, IT Technician	Learners
Backup validation	DR Technician 1	BC/DR Coordinator	Security Officer	BC/DR Coordinator
System restoration	DR Technician 1 and 2	BC/DR Coordinator	Application Owners	Business Coordinators
Stakeholder communication	Communications Lead	BC/DR Coordinator	Legal/Compliance	Learners, staff, regulators
Return to primary site	Facilities Manager	BC/DR Coordinator	DR Technicians, Academic Coordinator	All staff

4.3 Temporary Procedures — 72-Hour Site Outage

F1 — Teaching and Assessment:

The alternate facility provides 25 workstations from 10:00. Of 120 examination candidates, 40 require centre workstations and 80 use own devices. The 25 available workstations are insufficient for all 40 centre-dependent learners at once. The Academic Coordinator must arrange two supervised sessions: Session A (25 learners, 10:00–11:30), Session B (remaining 15 learners, 11:30–13:00). Alternatively, if paper-based examination materials are accessible, a single paper session for all 40 can run from 11:00. The 80 learners using own devices are directed to a separate supervised area with network access confirmed by 10:30.

Assessment integrity requires an invigilator ratio of no less than 1:30, identity verification at entry, sealed paper materials or system-locked browser assessment, and a signed attendance register. All completed scripts or submission confirmations must be stored in a locked container or sent to a designated secure email address before learners leave the room. The Academic Coordinator countersigns all records.

F2 — Fees and Finance:

A manual receipt book with pre-numbered, sequentially issued receipts is activated. Each transaction requires the Finance Officer's signature and a learner ID reference. The external payment provider's transaction log is downloaded every two hours via mobile internet as the reconciliation baseline. No wallet record is updated until FIN01 is restored and a supervisor has reviewed the manual log against the provider data.

F3 — Admissions and Registration:

A paper log captures applicant name, contact number, programme, document reference, and timestamp. Each entry requires the Admissions Officer's signature. No application is confirmed until REG01 is restored and the entry is verified against the paper log. Applicants receive a written or emailed acknowledgement stating that their application is received but subject to system confirmation.

F4 — Staff Payroll:

No payroll changes are processed during the outage without dual authorisation from the HR Manager and the Executive Sponsor. The last verified payroll file is held offline. If the bank submission deadline of 12:00 Day 1 approaches before HR01 is restored, the HR Manager submits the last verified file with a written note to the bank that amendments may follow. No duplicate instructions are issued.

F5 — Records and Certificates:

Certificate and record requests are logged by name, learner ID, and request type. No certificates are issued until FS01 is restored and the Records Officer confirms file integrity. Requesters receive an acknowledgement within the 48-hour internal commitment window.

Minimum staff required: BC/DR Coordinator, Academic Coordinator, two invigilators, Finance Officer, Admissions Officer, HR Manager, Records Officer, two DR technicians.

4.4 DR Runbook

Step	Action	Owner	Prerequisite	Completion Check	Contingency if Fails
1	Activate recovery network	DR Tech 1	None	Network ping to offsite DR environment successful	Contact ISP; use mobile internet as fallback for communication
2	Restore AUTH01 from B03	DR Tech 2	Step 1 complete	Authentication test with two test accounts	Escalate to BC/DR Coordinator; extend manual procedures
3	Run security and backup checks	DR Tech 1	Step 2 complete	Integrity checksums verified; no unexpected admin logon detected	Halt restoration; escalate to security officer; do not proceed to DB01
4	Restore DB01 from B01 + B02 chain	DR Tech 2	Step 3 complete	Database connectivity confirmed; row count matches checkpoint record	Fall back to B01 full only; document data-loss window; notify business owners
5	Restore LMS01	DR Tech 1	Step 4 complete	LMS01 login confirmed; test learner account accessible	Extend paper examination; notify Academic Coordinator

Step	Action	Owner	Prerequisite	Completion Check	Contingency if Fails
6	Restore FIN01	DR Tech 2 (parallel with Step 5)	Step 4 complete	Finance login confirmed; last transaction record matches checkpoint	Continue manual receipts; update reconciliation log
7	Restore REG01	DR Tech 1 (parallel with HR01)	Step 4 complete	Admissions search function confirmed	Continue paper log
8	Restore HR01	DR Tech 2 (parallel with REG01)	Step 4 complete	Payroll file accessible; employee count confirmed	Proceed with offline payroll file if deadline within 2 hours
9	Restore FS01	DR Tech 1 or 2	Step 2 and Step 3 complete	File listing matches B04 manifest; access control check passed	Queue certificate requests; do not issue until integrity confirmed
10	Business acceptance	BC/DR Coordinator with each function owner	Steps 5–9 complete	Each function owner confirms service is usable	Identify and resolve specific gaps before declaring recovery
11	Incident response trigger	BC/DR Coordinator	Any step — if unexpected admin logon or data anomaly detected	Security officer notified; investigation opened	Pause restoration of affected system; preserve evidence

Rollback procedure: If DB01 restoration produces a corrupt state, revert to the B01 full backup, document the exact failure point, and notify all function owners of the extended data-loss window. If LMS01 fails after DB01 is confirmed clean, the paper examination session continues and LMS01 is re-attempted before declaring failure to the Executive Sponsor.

Technology restoration versus business service resumption: Completing Step 10 confirms that systems are technically operational. Business service resumption is declared only when a function owner has verified that they can perform real transactions, not when screens appear functional. These are distinct outcomes.

4.5 Stakeholder Communication Matrix

Stakeholder	Channel	Message Content	Timing	Owner
Learners	Email, notice at alternate site	Disruption notice, exam arrangements	By 09:00 Day 0	Communications Lead
Staff	WhatsApp group, email	Activation status, reporting instructions	By 08:30 Day 0	BC/DR Coordinator

Stakeholder	Channel	Message Content	Timing	Owner
Executive Sponsor	Direct call, email	Activation confirmed, cost authority needed	By 08:15 Day 0	BC/DR Coordinator
External payment provider	Email	Notification of outage; request transaction log	By 09:00 Day 0	Finance Officer
Regulatory body (if applicable)	Email	Data availability status	Within 24 hours if data access is impaired	BC/DR Coordinator
Media	Prepared statement only via Communications Lead	Brief factual statement	If enquiries received	Communications Lead

Initial learner notice (147 words):

"Northbridge Skills Institute — Important notice to all learners.

Due to flooding affecting our main site, all on-site services are temporarily relocated to our alternate facility. The main site is currently inaccessible and will remain so for an estimated 72 hours.

Learners scheduled for the 11:00 examination today should report to the alternate facility entrance by 10:30. Please bring your student identification card. Learners requiring a centre workstation will be assigned to supervised sessions beginning at 10:00. Learners using personal devices should ensure they are charged and connected to mobile data before arrival.

All other services, including fee payments and admissions, remain available through manual procedures. Staff are on site at the alternate facility to assist.

Further updates will be sent by email and posted at the alternate facility entrance. We regret the disruption and will restore full services as quickly as possible."

Incident response trigger: If Step 3 of the runbook detects an unexpected administrator logon to the backup system, or if payment records show transactions absent from the restored database, the BC/DR Coordinator immediately notifies the security officer and pauses restoration of the affected system. Incident response procedures then run concurrently with continuity operations. The two teams share situational awareness through a joint log maintained by the BC/DR Coordinator.

Return to primary site: Return is authorised only after facilities management provides written clearance, a safety inspection is completed, and the DR team has confirmed that production systems can be restarted cleanly. New records created during the outage — manual receipts, paper logs, email applications — are entered into restored systems before production resumes, under dual supervision. The BC/DR Coordinator countersigns the final reconciliation record and the Executive Sponsor signs off on recovery closure.

Task 5: Recovery Schedule and Exercise

5.1 Baseline Warm Site Schedule

Start: 08:00. Two technicians available. Actions cannot be split. Two actions may run in parallel only when prerequisites are complete and both technicians are free.

Step	Action	Technician	Start	Duration (min)	Finish	Prerequisite
1	Recovery network	Tech 1	08:00	30	08:30	None
2	AUTH01 restoration	Tech 2	08:30	30	09:00	Recovery network
3	Security and backup checks	Tech 1	09:00	30	09:30	AUTH01 restoration
4	DB01 restoration	Tech 2	09:30	90	11:00	Security and backup checks
5	LMS01 restoration	Tech 1	11:00	45	11:45	DB01 restoration
6	FIN01 restoration	Tech 2	11:00	30	11:30	DB01 restoration (parallel with LMS01)
7	REG01 restoration	Tech 1	11:45	30	12:15	DB01 restoration (Tech 1 free after LMS01)
8	HR01 restoration	Tech 2	11:30	45	12:15	DB01 restoration (Tech 2 free after FIN01)
9	FS01 restoration	—	—	60	—	AUTH01 + security checks (both techs occupied)

FS01 scheduling note: FS01 prerequisites (AUTH01 and security checks) are complete at 09:30. However, both technicians are then engaged on DB01 (Tech 2 from 09:30) and subsequently on post-DB01 applications. The earliest either technician is free after DB01 completes is 11:30 (Tech 2, after FIN01). FS01 therefore starts at 11:30 and finishes at **12:30**.

Summary of restoration finish times:

Function	System Restored	Finish Time	Elapsed from 08:00
F1	LMS01	11:45	3 hours 45 minutes
F2	FIN01	11:30	3 hours 30 minutes
F3	REG01	12:15	4 hours 15 minutes
F4	HR01	12:15	4 hours 15 minutes
F5	FS01	12:30	4 hours 30 minutes

Dependency path controlling assessment service (F1):

Recovery network (30 min) → AUTH01 (30 min) → Security and backup checks (30 min) → DB01 (90 min) → LMS01 (45 min) = **225 minutes total = 11:45**

Comparison with objectives:

Function	Modelled Finish	RTO Target	RTO Met?	RPO (07:00 checkpoint)	RPO Met?	MTD	Within MTD?
F1	11:45	10:00	No	1 hour (07:00 is 1 hr before disruption)	Yes	11:00	No
F2	11:30	14:00	Yes	2 hours	Yes	16:00	Yes
F3	12:15	16:00	Yes	4 hours	Yes	18:00	Yes
F4	12:15	08:00 Day 1	Yes	8 hours	Yes	12:00 Day 1	Yes
F5	12:30	08:00 Day 1	Yes	24 hours	Yes	08:00 Day 2	Yes

These are modelled baseline results from drill timings, not a live test. They represent the best achievable outcome when all actions complete in exactly their estimated durations and no failures occur.

Critical finding: F1 (LMS01) does not restore before the 11:00 examination deadline under the warm site schedule. The continuity workaround — paper-based or supervised device-based examination — is mandatory for Day 0 regardless of which recovery package is selected. The RTO target of 10:00 for F1 is not achievable with two technicians following this action sequence. This is a structural constraint, not a resource or skill gap: DB01 restoration alone takes 90 minutes after a 90-minute prerequisite chain, placing LMS01 completion no earlier than 225 minutes after activation.

F2 through F5 meet their RTO and MTD targets under the baseline schedule.

5.2 Tabletop Exercise — 45 Minutes

Exercise purpose: To test decision-making and plan adaptability under compounding adverse conditions, using the baseline warm site scenario as the starting point.

Participants (all roles simulated): BC/DR Coordinator, Academic Coordinator, Finance Officer, HR Manager, DR Technician 1, DR Technician 2, Communications Lead.

Pre-exercise briefing (Minutes 0–10): The team reviews the baseline schedule and confirms the Day 0 situation: flooding confirmed at 08:00, two technicians deployed to the warm site, paper examination workaround activated, alternate facility open with 25 workstations.

Minute 10 Inject — Site closure extended from 72 hours to five days

Decision owner: BC/DR Coordinator, with Executive Sponsor authorisation.

Decision: Extend all manual temporary procedures through Day 5. Notify learners and staff of the extended outage. Escalate alternate facility agreement from a 72-hour arrangement to a five-day booking, confirming cost authority with the Executive Sponsor. Assess whether additional workstations can be sourced through equipment hire, and whether planned teaching sessions for Days 2 through 5 can be delivered via Microsoft Teams using mobile internet.

Rationale: A five-day outage extends beyond the 48-hour F5 service commitment and creates sustained pressure on F4 (payroll deadline is Day 1). F4's deadline is not affected by the extended outage itself, but any manual payroll file must still be submitted by 12:00 Day 1. Records and certificate requests accumulate; the queue must be communicated to requesters with a revised timeline.

Effect on service: Teaching continuity requires a remote delivery plan. Certificate and record commitments must be renegotiated with requesters beyond Day 2. Budget authority for the extended alternate facility hire must be confirmed within the hour.

Minute 20 Inject — Alternate facility reduces to 15 working workstations

Decision owner: Academic Coordinator, advised by BC/DR Coordinator.

Decision: Revise examination arrangement immediately. Of 40 learners requiring centre workstations, only 15 can be accommodated per session. Three sequential sessions are now required: Session A (15 learners, 10:00–11:30), Session B (15 learners, 11:30–13:00), Session C (10 learners, 13:00–14:00). Alternatively, switch to a fully paper-based examination for all 40 centre-dependent learners in a single session, which eliminates the workstation constraint entirely if paper materials are available.

Rationale: Running three sessions with rotating invigilators creates integrity risk: later-session learners may receive information from earlier-session learners. If paper materials are available, a single paper session removes this risk. If only digital materials exist, a strict no-communication protocol must be enforced with phones collected at entry, and sessions must not overlap in the facility.

Effect on service: The examination rescheduling cost of NGN 120,000 is already triggered. Extending sessions into the afternoon affects F2 and F3 staff who are managing manual procedures in the same facility. A separate room allocation for examination and administrative functions is required.

Minute 30 Inject — DB01 incremental backup fails integrity check; unexpected backup administrator logon detected

Decision owner: DR Technician 1 (technical), BC/DR Coordinator (escalation), Security Officer (investigation).

Decision — backup failure: Do not proceed with DB01 restoration using the broken incremental chain. Attempt to restore DB01 from B01 (full backup, 23:00 Day -1) only. This shifts the recoverable data point from 07:00 to 23:00 the previous evening — a nine-hour data-loss window instead of one hour. All function owners must be notified immediately of the extended RPO breach.

Decision — unexpected logon: Halt backup operations. Preserve the access log. Notify the Security Officer and open an incident response investigation concurrently with continuity operations. The offsite provider is contacted to verify whether the logon originated from their environment and to confirm whether any backup files were accessed or modified. Do not use backup files that may have been accessed by an unverified identity until integrity is independently confirmed.

Rationale: Proceeding with a potentially compromised backup risks restoring a corrupt or malicious state into the recovery environment. The cost of a clean restart from B01 — extended data loss, manual reconciliation — is preferable to propagating an unknown threat. This is the scenario in which incident response and BC/DR must operate simultaneously, as described in the course slides (ICDFA, 2025).

Effect on service: F2 RPO is now nine hours (23:00 checkpoint) against a two-hour target — a significant breach. Finance Officer must extend manual reconciliation to cover all transactions from 23:00 Day -1 to 08:00 Day 0 using provider logs and paper records. F4 data loss extends to nine hours; payroll file must be verified against the last confirmed HR01 state before submission. All function owners log additional manual records from 23:00 onward.

Minute 40 Inject — Payment provider reports successful payments absent from restored finance records

Decision owner: Finance Officer, escalated to BC/DR Coordinator and Security Officer.

Decision: Do not reconcile provider records against the restored DB01 state until the cause of the discrepancy is established. Two explanations are possible: the payments occurred between 23:00 Day -1 and 08:00 Day 0 and are absent because the recovery point is the 23:00 full backup; or the restored data is incomplete or corrupt in a way not caught by the initial integrity check. The Finance Officer cross-references provider timestamps against the 23:00 checkpoint boundary to determine which explanation applies.

Rationale: If the missing payments fall between 23:00 and 08:00, they are expected data-loss within the now-confirmed nine-hour window and can be manually reconciled from provider logs. If any missing payment timestamp predates 23:00, the DB01 restoration must be treated as incomplete, and the Security Officer must assess whether the logon event at Minute 30 caused data manipulation.

Effect on service: Fee confirmation cannot proceed until the discrepancy is resolved. The 16:00 F2 deadline is now at risk. Finance Officer extends manual receipting, holds all fee confirmations, and notifies affected learners that confirmations are delayed pending system verification. If the 16:00 deadline cannot be met, the Finance Officer contacts the relevant body — internal or regulatory — to document the delay and its cause.

5.3 Corrective Actions

Ref	Corrective Action	Priority	Owner	Target Date	Completion Evidence
CA-01	Separate backup administration identity from production administration. Create a dedicated backup service account with no production login rights.	Critical	IT Manager	30 days	Audit log confirms no shared credentials; access review signed off by Security Officer
CA-02	Obtain and document the offsite provider's access-independence controls, including evidence that backup files cannot be accessed using production credentials.	Critical	BC/DR Coordinator	30 days	Written confirmation from provider; independent verification test completed and recorded
CA-03	Reduce DB01 incremental backup interval from the current overnight gap to every two hours during business hours, closing the recoverable data-point gap from nine hours to two hours maximum.	High	IT Manager	60 days	Schedule confirmed in backup tool; two consecutive cycles verified with integrity checks

Ref	Corrective Action	Priority	Owner	Target Date	Completion Evidence
CA-04	Source and document an alternate examination delivery procedure that does not depend on system restoration, including pre-printed paper materials for all scheduled examinations and a confirmed invigilator roster.	High	Academic Coordinator	60 days	Paper examination kit maintained at alternate facility; procedure tested in next scheduled exercise
CA-05	Schedule and conduct a live restoration test of the full DB01 incremental chain, from B01 through the latest incremental, in an isolated environment. Record pass/fail result, restoration time, and data consistency check outcome.	High	DR Technician 1	45 days	Restoration test report signed by BC/DR Coordinator; result entered into KPI dashboard

Task 6: Governance and Performance Measures

6.1 Plan Ownership and Approval

Plan owner: BC/DR Coordinator — responsible for day-to-day maintenance, update initiation, and exercise coordination.

Plan approver: Executive Sponsor (CEO or delegated COO) — signs off on all major revisions, budget commitments, and strategy changes.

Review triggers:

- Scheduled annual review, minimum.
- Following any activation of the BC/DR plan.
- Following any significant organisational change: new systems, change in staff numbers above 10%, new premises, or new regulatory obligation.
- Following each test or exercise, based on corrective actions raised.

Version control: Each plan version carries a version number, effective date, and the name of the approving executive. Superseded versions are archived but not distributed. The current version is stored in Microsoft 365 SharePoint with access restricted to plan users, and a printed copy is held at the alternate facility.

Distribution: Controlled distribution list maintained by the BC/DR Coordinator. All recipients are notified when a new version is issued and confirm receipt by email.

Training: All staff with a named role in the plan complete an annual awareness session. DR technicians and function coordinators complete a procedural walkthrough semi-annually. New staff with plan roles are trained within 30 days of appointment.

6.2 Risk-Based Testing Calendar

Test Type	Frequency	Scope	Owner
Tabletop exercise	Quarterly	Varied scenarios; at least one flood or power event per year	BC/DR Coordinator
Walkthrough	Semi-annually	All five functions and RACI roles	BC/DR Coordinator
Backup restoration test	Every 45 days	Full DB01 chain and at least one incremental; FS01 full	DR Technician 1
Parallel test (warm site)	Annually	Full warm site activation without interrupting production	BC/DR Coordinator and DR Team
Full interruption test	Every 3 years	Complete failover to alternate site	Executive Sponsor authorised

6.3 Framework Mapping

NIST SP 800-34 alignment (NIST, 2010):

NIST Step	Assignment Output
Step 2 — Conduct BIA	Task 1: five-function BIA, impact calculations, priority sequence
Step 4 — Create Contingency Strategies	Task 3: cold, warm, hot comparison and budget recommendation
Step 5 — Develop ISCP	Task 4: DR runbook, temporary procedures, RACI, communication matrix
Step 6 — Ensure TT&E	Task 5: baseline schedule, tabletop exercise, corrective actions

ISO 22301:2019 alignment (ISO, 2019):

ISO 22301 Clause	Assignment Output
Clause 8 — Operation (BIA and BC strategies)	Tasks 1, 2, and 3: BIA, recovery objectives, strategy selection
Clause 8 — Operation (BC plans and procedures)	Task 4: coordinated plan, runbook, temporary procedures
Clause 9 — Performance evaluation	Task 6: KPIs, testing calendar, maturity assessment
Clause 10 — Improvement	Task 5: corrective actions CA-01 through CA-05

This mapping reflects educational alignment with published framework structures. It does not constitute or imply certification.

Flood and climate consideration:

ISO 22301:2019 Amendment 1:2024 introduces explicit requirements to consider climate-related risks within the BCMS context (ISO, 2024). For Northbridge, flooding is the confirmed disruption trigger in this scenario. The organisation's BIA and risk register should document flood probability based on the site's location, drainage infrastructure, and regional climate data. The alternate facility must be verified

as outside the same flood zone as the primary site — an alternate facility that floods simultaneously provides no protection. The BC/DR Coordinator should obtain flood risk assessments from the relevant municipal authority and factor seasonal rainfall patterns into the testing calendar, scheduling at least one flood-scenario tabletop during the wet season each year.

6.4 Key Performance Indicators

KPI	Formula / Measurement Method	Data Source	Target	Frequency	Accountable Owner
Recovery time performance	(Target RTO − Actual RTO achieved in test or live event) / Target RTO × 100. Positive value means target met.	DR runbook completion timestamps, test reports	100% of functions meet RTO in scheduled parallel test	After each test or activation	BC/DR Coordinator
Backup restoration test coverage	(Number of backup chains successfully restored and verified in period) / (Number scheduled) × 100	Backup test log	100% of scheduled tests completed and passed	Monthly	DR Technician 1
Corrective action closure rate	(Number of corrective actions closed by target date) / (Total corrective actions raised) × 100	Exercise log and corrective action register	90% closed on time; 100% closed within 30 days of target	Quarterly	BC/DR Coordinator
Training completion rate	(Staff with plan roles who completed required training in period) / (Total staff with plan roles) × 100	Training attendance records in Microsoft 365	100% completion before annual review	Annually, with quarterly progress check	HR Manager

6.5 Maturity Assessment and Improvement Target

Current maturity — Level 2: Developing (Repeatable)

The evidence from this scenario supports a Level 2 assessment. Basic plans exist or are now being developed for critical systems. The warm site option provides a defined recovery path. However, backup integrity has not been independently tested, access-independence controls are undocumented, the backup administrator identity is shared with production, and no regular testing calendar exists. Recovery

depends on two technicians executing a sequence they have not fully rehearsed. These are characteristics of a programme that is repeatable in favourable conditions but inconsistent under stress.

12-month improvement target — Level 3: Defined (Standardised)

Within 12 months, the programme should reach Level 3 by completing the five corrective actions from Task 5, establishing the testing calendar in Section 6.2, delivering two tabletop exercises and one walkthrough, completing at least eight backup restoration tests, and separating backup administration credentials. Level 3 is confirmed when all five functions have tested procedures, governance is documented and approved, and KPI results are reported to the Executive Steering Committee at least twice in the year.

The step from Level 2 to Level 3 does not require new technology investment beyond what the warm package already funds. It requires consistent execution of the processes defined in this plan, with results recorded and acted upon.

References

ICDFA. (2025). *Business continuity and disaster recovery*. GRC205 — Incident Management and Business Continuity, Week 21 teaching deck, slides 1–15. International Cybersecurity and Digital Forensics Academy.

ISO. (2019). *ISO 22301:2019: Security and resilience — Business continuity management systems — Requirements*. International Organization for Standardization. https://www.iso.org/standard/75106.html

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NIST. (2010). *SP 800-34 Rev. 1: Contingency planning guide for federal information systems*. National Institute of Standards and Technology. https://csrc.nist.gov/publications/detail/sp/800-34/rev-1/final

All scenario data, organisations, names, and figures in this report are fictional and created for assessment purposes only. AI assistance was used in drafting and structuring this report in accordance with course policy.